A RAG Pipeline for Phenotype Extraction and HPO Normalization

Toward rare-disease diagnosis support

Vania Raya

Dynamic and Distributed Information Systems Group
University of Zurich

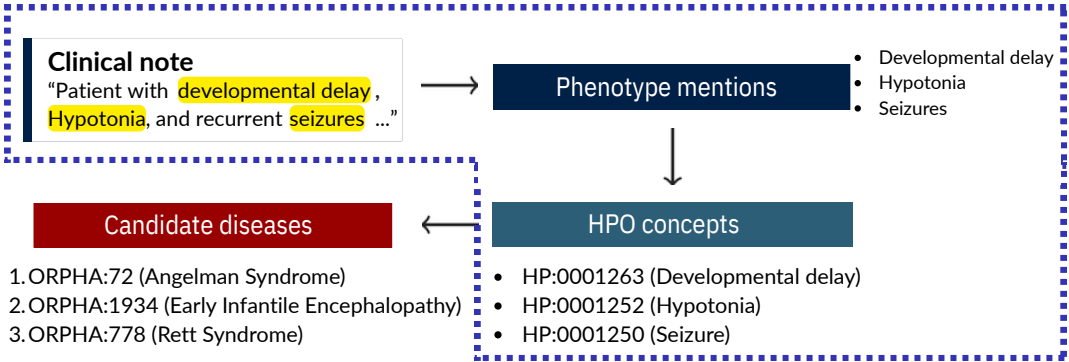
Supervisors: Dr. Selene Báez Santamaría & Prof. Abraham Bernstein

August 2026





Rare-disease diagnosis depends heavily on phenotype extraction



Impact ~500,000 people in Switzerland live with a rare disease

Objective To build a pipeline that converts clinical notes into an HPO phenotype set

Evaluating retrieval for phenotype-to-HPO mapping

Research question

Can a RAG system improve phenotype extraction and HPO normalization from clinical text?

Hypotheses

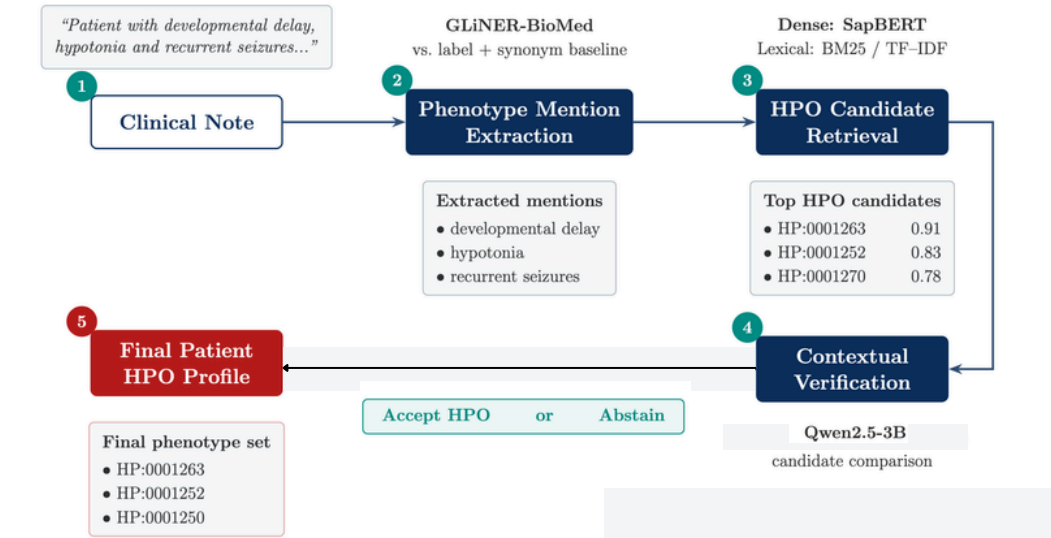
- H1: Semantic retrieval yields higher candidate recall than lexical matching
- H2: An LLM verifier improves precision compared to raw retrieval
- H3: Integrating the strongest extractor and retriever improves end-to-end HPO quality

Existing systems already integrate RAG but precise distincion among close HPO concepts remains challenging

System	Phenotype Extraction	HPO Retrieval	Uncertainty / Abstention	Ontology-aware Error Analysis
Remo / UZH Wiget, 2025	✓	✓	✗	✗
RAG-HPO 2025	✓	✓	∼	✗
AutoPCR 2026	✓	✓	∼	∼
RARE-PHENIX 2026	✓	✓	✓	∼
This work UZH, 2026	✓	✓	✓	✓

✓ explicitly addressed ∼ partially addressed ✗ not a primary component

System architecture: clinical text to HPO profile



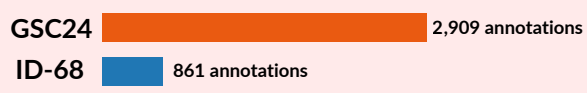
Datasets stress different parts of the system

Dataset	Size	phenotypes/doc	Direct lexical match
ID-68	68 documents	11	~ 40%
GSC24	228 documents	11	~ 20%

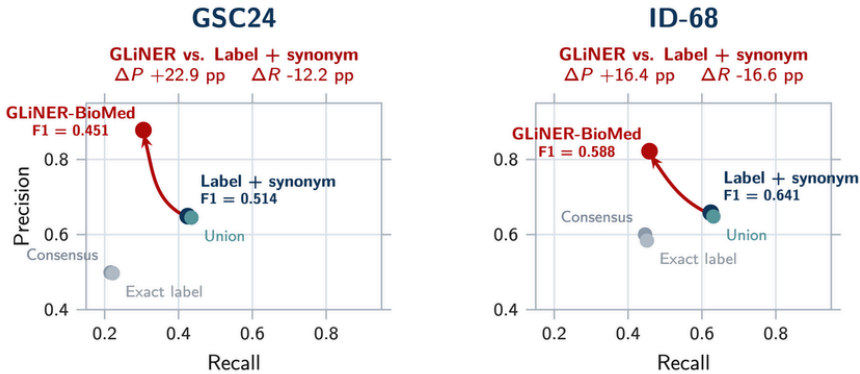
Precision/Recall/F1/MRR
Extraction and HPO retrieval metrics

Uncertainty estimation
95% bootstrap CI

GSC24 is more challenging because mentions significantly deviate from canonical HPO terminology.

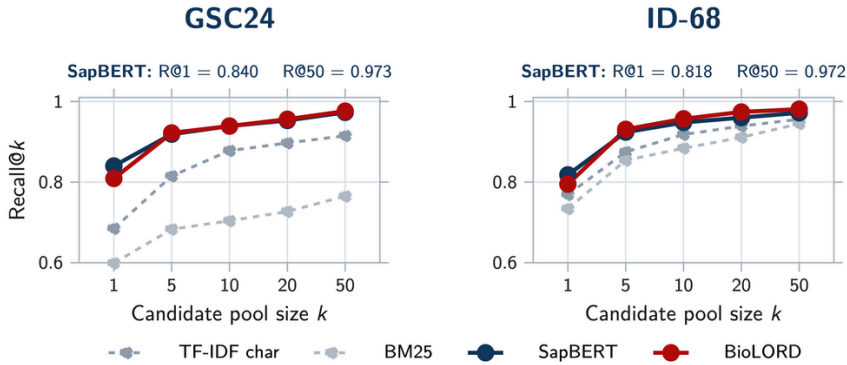


Extraction exposes a precision–recall trade-off, not a single best model



GLiNER greatly increases precision, but misses more phenotypes; extraction remains the dominant recall bottleneck.

Dense retrieval already finds the correct entity-linking HPO concept surprisingly well



Once a phenotype mention is available, candidate matching is not the main problem.

Generic rerankers are not enough for HPO disambiguation

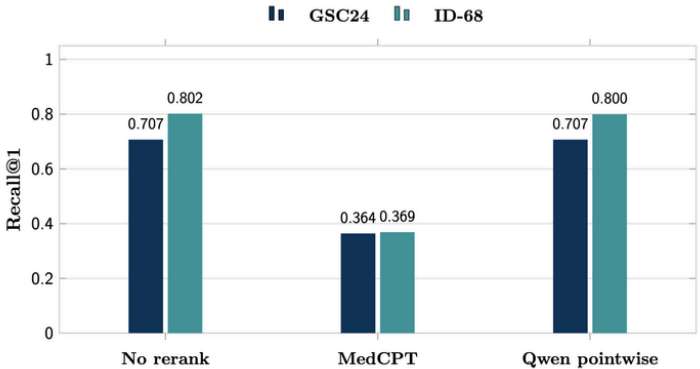
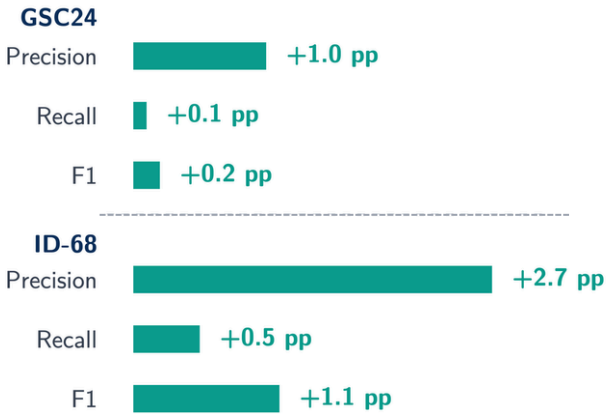


Figure 1: Recall@1 results for candidate models.

The zero-shot cross-encoder actively harms ranking; the pointwise LLM leaves the original order unchanged.

All three methods rerank the same R5 SapBERT+lexical pool.

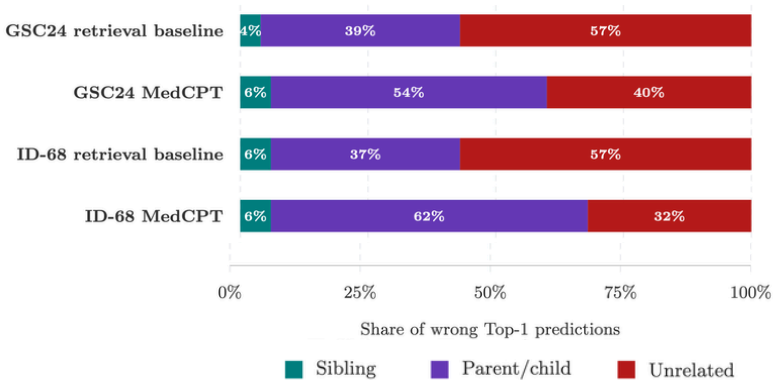
The full pipeline yields modest gains beyond a strong GLiNER baseline



Most end-to-end performance comes from the extractor.

Retrieval and verification provide **small but consistent gains**, especially in **precision**.

MedCPT successfully restricts candidates to the correct clinical domain



What changes?

MedCPT reduces the fraction of **completely unrelated HPO errors**:

GSC24
57% → 40%

ID-68
57% → 32%

Main takeaways

1. **Phenotype extraction** remains the largest source of error in the pipeline.
2. **Dense HPO retrieval** is highly effective at recovering correct candidates.
3. **Generic rerankers** are insufficient for fine-grained HPO disambiguation.
4. **Traditional baselines** suffer from a 57% failure rate, mapping to irrelevant branches of medicine

Conclusion:
The main bottlenecks lie in phenotype extraction and fine-grained HPO disambiguation, while candidate retrieval is already strong.

Limitations and future directions

⚠ Current Limitations

1. **Limited extraction recall**
2. **Verifier contribution unresolved**
3. **No German clinical validation yet**

⋙ Next Steps

1. **Stronger hybrid extraction**
2. **HPO-specific listwise verifier**
3. **Calibrated abstention + German evaluation**

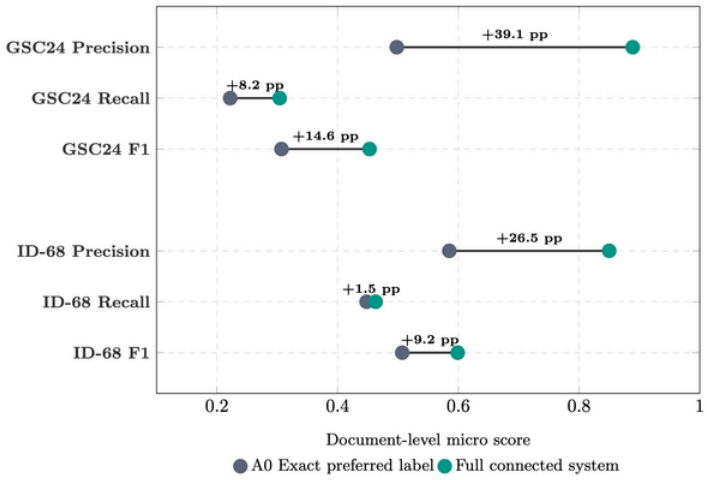
> The next gains are unlikely to come from broader retrieval, but from better extraction, fine-grained HPO disambiguation, and calibrated uncertainty.

Full ablation matrix

Component	Variant	Metric	GSC24	ID-68	Evidence
Extractor	Exact label (A0)	Doc F1	0.307	0.507	Weak baseline
Extractor	+ exact synonyms (A1)	Doc F1	0.514	0.641	Large gain
Extractor	GLiNER + A1 linker	Doc F1	0.451	0.588	Precision trade-off
Retriever	SapBERT	R@1 / R@50	0.840 / 0.973	0.818 / 0.972	Strong
Reranker	MedCPT zero-shot	R@1	0.364	0.369	Hurts
Reranker	Qwen pointwise	R@1	0.707	0.800	No effect
Full system	GLiNER + retrieval + verifier	Doc F1	0.453	0.599	Modest gain

> the pipeline is candidate-rich but extraction-limited; off-the-shelf rerankers do not bridge the gap.

The full system substantially improves precision over exact matching



Most of the improvement comes from precision, while recall gains remain comparatively modest.

AutoPCR system

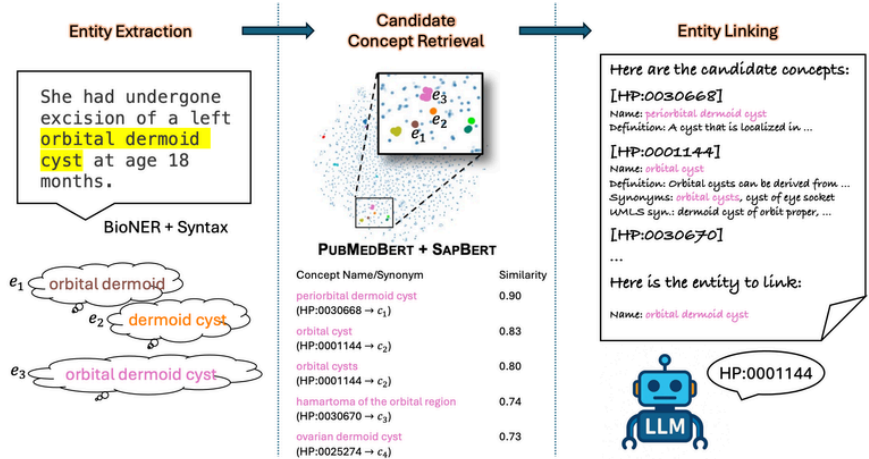


Figure 1 Architecture of AutoPCR. It performs concept recognition in three stages: entity extraction using BioNER and syntactic entity extraction (e.g., extracted entities e_1, e_2, and e_3), candidate concept retrieval via SapBERT initialized from PubMedBERT (e.g., retrieved concepts c_1, c_2, c_3, and c_4 for e_3), and entity linking through prompting an LLM (e.g., linked concept c_2).

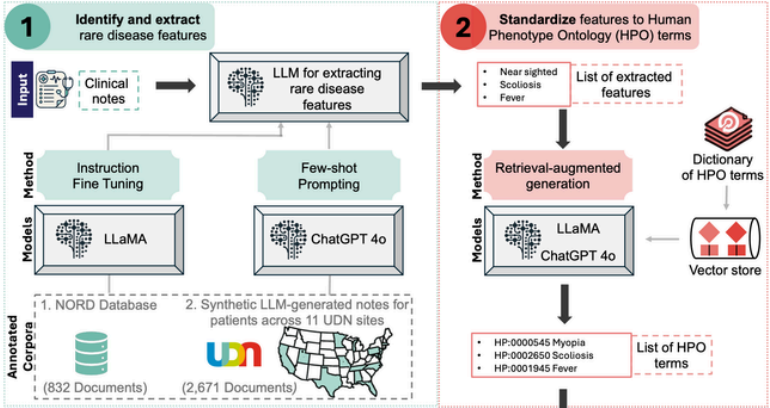
AutoPCR system

Contribution: Adapts instantly to new or updated biomedical ontologies

Limitations:

- **Ignores Clinical Uncertainty**: Treats speculative text (e.g., "possible" or "rule out") as absolute, definitive conditions.
- Sequential three-stage architecture limits s**caling speed and computational throughput**.
- Validation was restricted to relatively small, highly domain-specific medical corpora.

RARE-PHENIX system



AutoPCR system

Contributions: Bridges the operational gap by executing phenotype extraction, HPO standardization, and diagnostic prioritization in one workflow.

- Employs specialized, parameter-efficient LoRA models fine-tuned directly on extensive rare-disease corpora.
- Filters clinical noise by using an XGBoost ranking model to surface highly informative phenotypes over generic symptoms.

Limitations:

- Achieving peak performance requires massive **70B parameter models** that are difficult to host locally.
- Lacks real-world validation tracking actual downstream impact on clinician workloads.